

Tutorial 9 — Exercise 1: Deriving the Difference-in-Differences Estimator

Introductory Econometrics — TA Session

Duration: 50 minutes | Based on Lecture 16 (Marmer)

Lecture 16 introduced the DID regression:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

and stated that under parallel trends and no anticipation, $\beta = \text{ATT}$. In this exercise you will derive *why* this is true, step by step, using potential outcomes.

Part 1: Setup

For each individual i in each period t , define two **potential outcomes**:

- $Y_{it}(0)$: the outcome if individual i is **not treated** in period t .
- $Y_{it}(1)$: the outcome if individual i is **treated** in period t .

Treatment is active only for $D_i = 1$ at $t = 1$. The observed outcome is:

$$Y_{it} = D_i \cdot t \cdot Y_{it}(1) + (1 - D_i \cdot t) \cdot Y_{it}(0).$$

(a) Verify that this gives the following table of what we observe:

	Control ($D_i = 0$)	Treatment ($D_i = 1$)
$t = 0$	$Y_{i0}(0)$	$Y_{i0}(0)$
$t = 1$	$Y_{i1}(0)$	$Y_{i1}(1)$

The **ATT** is $E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$. We observe $Y_{i1}(1)$ for the treated, but we **never observe** $Y_{i1}(0)$ for them — the missing counterfactual.

Part 2: DID in potential outcomes

(b) Write the DID estimand $\beta = E[Y_{i1} - Y_{i0} \mid D_i = 1] - E[Y_{i1} - Y_{i0} \mid D_i = 0]$ in terms of potential outcomes, using part (a).

Part 3: Connecting DID to the ATT (key derivation)

(c) Start from part (b). Inside the first expectation, **add and subtract** $Y_{i1}(0)$ and $Y_{i0}(1)$. Rearrange to show:

$$\beta = \underbrace{E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]}_{\text{difference in trends}} + \underbrace{E[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1]}_{\text{anticipation effect}}.$$

Hint: This is the decomposition from Lecture 16, slide 3. Add zero in a clever way, then regroup.

Part 4: The two assumptions

(d) **Parallel trends.** State the assumption that kills the “difference in trends” term. Show it does *not* require equal levels across groups.

(e) **No anticipation.** State the assumption that kills the “anticipation effect” term. Give an example of when it might fail.

(f) Combine parts (c)–(e). Show $\beta = \text{ATT}$. State which assumption eliminates which term.

Part 5: From DID to the regression

(g) Recall the DID regression from Lecture 16:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it}, \quad E[U_{it} \mid D_i] = 0.$$

Evaluate $E[Y_{it} \mid D_i]$ at each of the four cells. Then compute the double difference and show it equals β .

Part 6: What goes wrong when parallel trends fails

(h) Suppose no anticipation holds but parallel trends does not. Show that $\beta = \text{ATT} + B$, where:

$$B = E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0].$$

Give an example where $B > 0$ and one where $B < 0$.

Part 7: Numerical verification

Consider $n = 4$ individuals. The column $Y_{i1}(0)$ is **unobserved** for treated units; shown only for verification.

i	D_i	Y_{i0}	Y_{i1}	$Y_{i1}(0)$	$Y_{i1}(0) - Y_{i0}$
1	1	5	12	8	3
2	1	7	15	10	3
3	0	4	7	7	3
4	0	6	9	9	3

(i) Verify parallel trends. Compute the true ATT. Compute DID from observed data only and confirm $\hat{\beta} = \text{ATT}$.

(j) Read off the OLS coefficients $\hat{\alpha}, \hat{\delta}, \hat{\gamma}, \hat{\beta}$ from the cell means and verify against the 2×2 table.